# Detecting kidney stones

Sometimes substances that are normally dissolved in urine separate out as a solid, making hard little stones called **kidney stones**. The stones move through the excretory system, scratching and blocking the thin tubes of the kidney, ureter and urethra. It is very painful. There are three ways for doctors to see if a patient has kidney stones.

**Standard X-ray –** low doses of **radiation**

- Uses a photographic negative
- Detects bones and calcium stones
- Cheap and quick

**Ultrasound** – sound waves passed over the skin near kidneys, ureters and bladder

- Image presented on screen
- Detects both calcium and non-calcium stones
- No radiation

**CAT scans** – an X-ray tube that creates many images

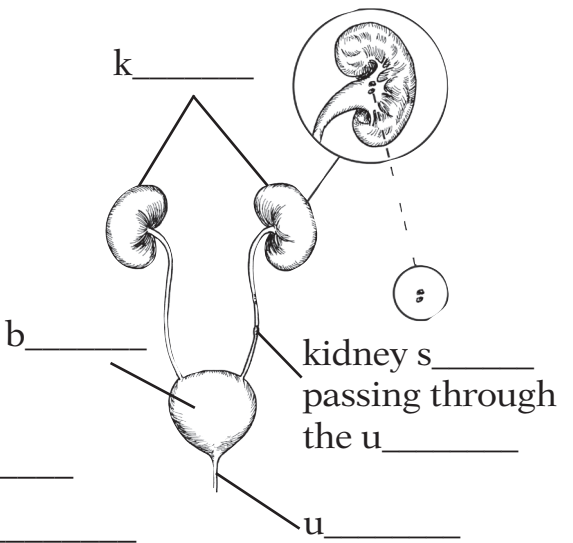
- Sometimes detects non-stone causes of severe pain
- Sensitive but expensive
- Uses more radiation than a conventional X-ray.

☞ **1** Which method is best for someone who shouldn't be exposed to radiation, such as a pregnant woman? _____

**2** Which method is best for someone in severe pain? _____

**3** Use the word bank to label the diagram.

| Word bank | | | | |
|---|---|---|---|---|
| kidneys | ureter | stones | urethra | bladder |




**4** Are these statements true or false?

**a)** Kidney stones are only found in the kidney. _____

**b)** A standard X-ray is cheap and quick. _____

**c)** CAT scans can only detect kidney stones. _____

**d)** An ultrasound does not use radiation. _____


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